Optimal Power Flow: A Literature Review

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***Abstract*—This paper presents a literature review on optimal power flow (OPF) for the analysis, planning, and effective operation of electric power systems. It compiles the description, formulation, and selected applications of the main OPF models, namely the DC and AC models, together with the most common computational tools used to solve them, given their characteristic complexity. The review shows how the different models are applied across a wide range of problems that require power-flow calculations, reflecting their importance in the optimization of electric power systems.**

***Index Terms*—Literature review, optimal power flow (OPF), computational tools for OPF, electric power systems (EPS).**

I. Introduction

An electric power system (EPS) can be divided into three main stages: generation, transmission, and distribution. The electrical energy produced by the generators is carried through the transmission system and delivered to customers through the distribution network [1]. The EPS is a highly complex system because of its size and the uncertainty of several variables: it is not static or deterministic but, in practice, stochastic. Every country has its own EPS, managed by an operator whose task is to maintain the real-time energy balance of the network while guaranteeing reliability and safe operation. Because load behavior is not fixed or controllable, the system must adapt to it, which makes control more complex.

Modern power systems have adapted to the massive inclusion of renewable technologies—mainly photovoltaic and wind—as well as battery energy storage systems [2]. However, the system tends to change its behavior over time: energy demand grows inevitably, and the exploitation of renewable and non-renewable resources, together with rising raw-material and generation costs, makes real-time operating tools necessary to guarantee optimal operation.

Ensuring safe and efficient operation requires algorithms and methodologies that provide fast and robust solutions [3]. This is where optimization problems such as economic dispatch (ED) come in. ED enforces the system balance by finding the combination of units—and the amount of energy each unit must dispatch—to supply the load, with the objective of economical operation. A more complex but more accurate problem is optimal power flow (OPF), which is more precise because it includes losses and an exact model of the system. The two problems are not unrelated: OPF is closely tied to the ED formulation, since both rely on the power-flow calculation, so OPF can be regarded as an evolution or extension of ED.

TABLE I
Variables of the OPF problem.

Variable	**Type**
State variables	
Bus voltage magnitude	Continuous
Bus voltage angle	Continuous
Real and imaginary parts of bus voltage	Continuous
Bus active power	Continuous
Bus reactive power	Continuous
Control variables	
Generated active and reactive power	Continuous
Transformer TAP position	Discrete
Phase-shifting transformer phase angle	Cont., Disc.
FACTS (compensation systems)	Cont., Disc.

ED and OPF are formulated and solved through optimization models, which have three key parts: the mathematical expression to be optimized, the variables, and the constraints. The model returns the settings that make the variables optimal for stable operation, meaning the variable values must satisfy the stated constraints. Table I, adapted from [1], lists the variables most frequently found in OPF problems.

According to [3], the constraints can be summarized in two sets: the physical limits and the operational limits of the system. The physical limits enforce the active- and reactive-power balance at each node and the dynamics of the power flows over the interconnections; they are defined by equality relations. The operational limits are defined by inequality relations that represent the operating bounds of each element, such as the thermal limits of lines and transformers, generator capability curves, permissible voltage bands, and the voltage-angle reference.

Finally, there is the objective function. There is no single objective function for OPF, since studies focus on different aspects—economic, environmental, or security related. Depending on the objective function and the constraints, the OPF problem admits different mathematical formulations, broadly classified as follows [4]:

- Linear problems with continuous control variables.
- Nonlinear problems with continuous control variables.
- Mixed-integer linear problems with discrete and continuous control variables.

Table II compiles objective functions from several studies [1], [2], [5].

TABLE II
COMMON OBJECTIVE FUNCTIONS FOR OPF OPTIMIZATION MODELS.

Objective function
Minimization of generation costs
Minimization of network losses
Minimization of active-power losses
Minimization of investment in reactive-compensation devices
Optimal voltage profile
Integration of renewable sources
Maximization of power quality
Minimization of greenhouse gases
VAR planning
Congestion
Transmission
Stability
Flexibility
Continuity

Because of the number of parameters and variables, OPF is solved on computers; the main challenge for the research community is to obtain quality solutions with short processing times—that is, to be more efficient and effective. This need for faster solutions has driven the development of several computational tools over time [6]. This paper reviews OPF under its different interpretations, covering both the formulation and the computational solution techniques, to provide an updated overview of EPS optimization while deepening the understanding of the system itself.

II. DC OPTIMAL POWER FLOW (DC OPF)

A. Conventional DC OPF

DC optimal power flow (DC OPF) is widely used in electrical engineering because it is an optimization problem with a simpler formulation and a fast solution [6]. It is essentially a linearization of the AC power flow that assumes the following [7]:

- Line resistance is very small compared with reactance and is therefore neglected.
- The voltage at every bus is equal to 1 p.u.
- The voltage-angle difference between adjacent nodes is small.

The first assumption implies that losses are not considered. The popularity of the method lies in providing a good approximation to the AC power flow while being faster and easier to understand. DC OPF can determine supplier needs for energy sales [5] or the amount to dispatch to meet a demand; it focuses only on real (active) power and ignores losses.

The formulation starts by expressing all components—resistances, reactances, powers, and voltages of buses, generators, transformers, lines, compensators, and loads—in per unit. The admittance matrix is then built and the flow directions are set. The objective function is given in (1); although OPF is applied to many objectives, the minimization of generation costs is among the most common.

$$\text{O.F.} = \text{Min} \sum_{i=1}^{N_{Gen}} C_i \, P_{gen_i} \tag{1}$$

The main variable is the power to be dispatched. The most important constraint in any OPF is the power balance: the total power generated at a bus plus the power flowing between that bus and its neighbors must equal the power demanded at that bus. Starting from the full AC line-power-flow equations and applying the DC assumptions (negligible conductance, a flat 1 p.u. voltage profile, and $\sin\delta_{ij} \approx \delta_{ij}$), the active-power flow through a line reduces to (2).

$$P_{ij} = \frac{\delta_i - \delta_j}{X_{ij}} = B_{ij}\left(\delta_i - \delta_j\right) \tag{2}$$

where δ_i are the bus voltage angles and X_{ij} is the line reactance. The complete power-balance constraint, requiring that generation plus flow equal demand, is given in (3).

$$P_{gen_i} - \sum_{j \neq i}^{J} \frac{\delta_i - \delta_j}{X_{ij}} = P_{load_i}, \quad \forall i \in N_{Bus} \tag{3}$$

The slack-bus constraint (4) sets the reference bus, with respect to which voltages are measured and through which currents find a return path.

$$\delta_{Slack} = 0 \tag{4}$$

The generation-limit constraint (5) bounds the power a unit connected to a bus can produce.

$$P_{gen_i}^{min} \leq P_{gen_i} \leq P_{gen_i}^{max}, \quad \forall i \in N_{Gen} \tag{5}$$

Finally, the line-capacity constraint (6) limits the active power the line can carry between two buses to known thresholds.

$$P_{ij}^{min} \leq \frac{\delta_i - \delta_j}{X_{ij}} \leq P_{ij}^{max}, \quad \forall i,j \in N_{Bus};\ i \neq j \tag{6}$$

An interesting study is presented in [9], which proposes two linearizations addressing the non-convexity of the DC OPF model through sequential quadratic programming and compares them with a nonlinear algorithm. The two proposals focus on linearizing the Jacobian stage of the Newton–Raphson method and on using nodal currents as auxiliary variables to linearize the inequality constraints; the results show the efficiency of the methodology and a close match with the exact nonlinear representation. Beyond economic dispatch, DC OPF is also used to analyze transmission-line loadability, voltage angles, and losses. For example, [10] addresses transmission-expansion planning applying contingencies, network switching, and line capacities based on the ideal SIL using DC OPF.

B. DC OPF Considering Losses

In power-system operation, locational marginal prices provide important economic indicators of the market operation, and they are commonly computed through a DC OPF that includes system losses [11]. This method keeps some of the conventional DC OPF assumptions, so it can be modeled as a linear program similar to the conventional one—minimizing generation costs subject to the balance, generation-capacity, and transmission constraints. Its advantage is that, since the conventional model already approximates the AC model, including losses yields even more accurate results.

Retaining the resistive terms, combining the sending- and receiving-end line flows, assuming 1 p.u. voltages, and using the identity $1-\cos\delta_{ij} = 2\sin^2(\delta_{ij}/2)$, the active-power losses on a line are approximated by (7).

$$P_{L_{ij}}^{LOSS} = 4\,G_{ij}\left(\sin\left(\frac{\delta_{ij}}{2}\right)\right)^2 \tag{7}$$

Since the angular difference is small, the power-balance constraint including losses can be written as (8).

$$P_{gen_i} - \left[\sum_{j=1}^{N_{Bus}} B_{ij}(\delta_j - \delta_i) + \sum_{j=1}^{N_{Bus}} 4\,G_{ij}\left(\frac{\delta_i - \delta_j}{2}\right)^2\right] = P_{load_i} \tag{8}$$

This loss analysis follows [6] and [11], but other loss approximations exist. In [12], an extension of the DC power flow that includes resistive losses is proposed, reformulating the equations as a fixed-point equation to obtain better loss approximations; the tests show fast convergence and a notable, theoretically supported gain in accuracy for radial networks. In [13], another line-loss model for use in linearized formulations such as DC OPF is proposed and compared with other methods, showing higher accuracy and computational advantages.

III. AC Optimal Power Flow (AC OPF)

The full or AC optimal power flow (AC OPF) does not make the strict approximations of DC OPF; instead, it describes the relation between injected power, voltage magnitude, and voltage angle at each bus through a set of nonlinear, nonconvex equations [14]. The model can also include discrete variables such as transformer TAPs, reactors and capacitors, and phase-shifting transformers that can be switched in or out of service [15]. According to [6], an adequate representation of the power flow at each bus yields a more efficient and accurate OPF model; the power-flow analysis is modeled on the Newton–Raphson method, which facilitates the determination of the decision variables.

Unlike the previous methods, AC OPF considers both active and reactive power, which changes the constraints. The active- and reactive-power balances are given in (9) and (10), representing the difference between the power produced by the generators and the power demanded by the loads at each bus.

$$P_{gen_i} - P_{load_i} = P_{flow_i}, \quad \forall i \in N_{Bus} \tag{9}$$

$$Q_{gen_i} - Q_{load_i} = Q_{flow_i}, \quad \forall i \in N_{Bus} \tag{10}$$

The power-flow terms are expanded in (11) and (12), where the influence of the Newton–Raphson power-flow analysis is evident; the flow depends on the bus voltage magnitudes and angles.

$$P_{flow_i} = \sum_{j=1}^{N_{Bus}} |V_i||V_j|\left(G_{ij}\cos(\delta_{ij}) + B_{ij}\sin(\delta_{ij})\right) \tag{11}$$

$$Q_{flow_i} = \sum_{j=1}^{N_{Bus}} |V_i||V_j|\left(G_{ij}\sin(\delta_{ij}) - B_{ij}\cos(\delta_{ij})\right) \tag{12}$$

The active-generation-limit and slack-bus constraints remain as in (4) and (5). The reactive-generation limit is given in (13).

$$Q_{gen_i}^{min} \le Q_{gen_i} \le Q_{gen_i}^{max}, \quad \forall i \in N_{Gen} \tag{13}$$

Other constraints common to AC OPF are the bus voltage-magnitude limits, the voltage-angle limits, and the transmission limits, given in (14), (15), and (16), respectively.

$$V_i^{min} \le V_i \le V_i^{max}, \quad \forall i \in N_{Bus} \tag{14}$$

$$-180^\circ \le \delta \le 180^\circ \tag{15}$$

$$S_{ij}^{min} \le \left| V_i^* \sum_{\substack{j=1 \\ i\in j}}^{N_{Bus}} V_j\,Y_{ij} \right| \le S_{ij}^{max} \tag{16}$$

As (16) shows, AC OPF accounts for the losses along the transmission lines through the relation between voltage and the admittance matrix. Although this method is more complex than the previous ones, it is worth developing so that simulations, studies, and results are as realistic as possible. The DC OPF literature focuses on general studies of power systems, whereas the AC OPF literature addresses more specific aspects, given the detail and precision of the model.

For example, [16] explores renewable-energy integration, emphasizing its variability and intermittency—a challenge for power flows—and examines innovative AC OPF strategies to guarantee economical and stable operation, while introducing new directions such as storage integration, linearization techniques, and data-based strategies. In [17], the constraints of a traditional AC OPF are linearized through Taylor series to obtain an accurate, applicable model that minimizes losses in transmission-system expansion, identifying which lines to replace or reinforce at minimum cost and losses. In [18], an extended OPF representing static VAR compensators (SVC) is formulated using sequential quadratic programming and decision-making methods such as those of Zoutendijk and Rosen; the results show that the proper introduction of these FACTS devices in a transmission network leads to lower total costs through active-power generation savings.

TABLE III
OPF SOLUTION METHODS.

Deterministic methods	Non-deterministic methods
Linear programming	Genetic algorithms
Nonlinear programming	Particle swarm
Gradient	Simulated annealing
Newton	Tabu search
Quadratic	Ant colony optimization
Simplex	Differential evolution
Interior point	Artificial neural network

IV. Computational Tools for OPF Analysis

Solving an OPF requires handling a large volume of data and complex optimization algorithms, which demands computational tools. The methods vary with the mathematical formulation, so there is no single way to solve these problems. According to [2], [4], OPF is addressed through three main routes: conventional exact mathematical-optimization methods, intelligent-search or metaheuristic optimization methods, and a non-quantitative approach for handling uncertainty in the objectives and constraints. Metaheuristics use algorithms inspired by nature or physics to determine the optimal power outputs of distributed generators, which are then used as inputs to conventional power-flow methods, whereas conventional analysis restructures the power-balance constraints into convex equivalents to guarantee that the global optimum is found.

Table III compiles the most common solution methods reported in [1], [19], [20]. A more detailed analysis of these methods is given in [19], [20], [21]. As an example, [22] presents an OPF solution for DC networks combining a Whale Optimization Algorithm (WOA) with three other algorithms, achieving greater power-loss minimization than the alternatives.

From the literature [1], [3], [6], two types of computational tools can be distinguished: commercial tools and programmable (open-source) tools.

A. Programming Languages and Libraries

These tools are open-source and easily accessible. Some of the most widely used are described below.

Pyomo. An open-source Python library for modeling different optimization problems. It uses the "Ipopt" solver to optimize nonlinear systems through Newton-type approximations and others.

Matpower. A Matlab-based tool developed specifically to solve power flow and optimal power flow. It ships with standardized IEEE test cases and solves PF and OPF (DC and AC). It is one of the pioneering tools for these problems and the basis for others such as Pypower, and it presents the optimization data in a tabular, structured form.

Pypower. A Python port of Matpower for power flow and optimal power flow. It solves AC and DC power flow with the Newton–Raphson, fast-decoupled, and (depending on the version) Gauss–Seidel methods. OPF, both AC and DC, is solved by default with the interior-point method, and DC OPF uses linear programming. Like other tools, it includes default IEEE test systems; it is the simplest of the Python tools, although its effectiveness is limited because it cannot model complex networks.

Pandapower. A power-system analysis library based on the "Pandas" data library and on Pypower. It solves OPF problems by configuring data, constraints, and parameters through Pandapower data structures, using either AC or DC power-flow equations. Its constraints include several of those presented in this work.

PowerModels. Arising from the popularity of the Julia language and its JuMP optimization package, this package focuses on solving different optimization problems through various formulations and approximations.

PFNET. A project for analyzing and modeling electrical networks. Its aim is to create optimization problems at a higher level than mathematical-modeling tools while offering more formulation detail than Matpower. However, for networks with a larger number of nodes, PFNET exhibits convergence problems and longer solution times, because it uses a simple implementation of the interior-point method for OPF.

B. Specialized Power-System Analysis Software

These tools are more specialized and do not focus solely on OPF; they also simulate cases to analyze the behavior of the power system.

DIgSILENT PowerFactory. A power-system analysis program used in generation, transmission, distribution, and industrial studies. The standard power flow first computes branch flows and bus voltages from defined "setpoints," and the OPF then finds the optimal values that satisfy the constraints and meet the objective. Its OPF options include minimization of network losses and costs, AC and DC flow optimization, control of generator active and reactive power, TAP positions, and SVS systems, together with the constraints presented above for both formulations.

ETAP. Offers a wide variety of specialized modules for power-system analysis and stands out for the ease of configuring electrical models. It provides a module to solve load flows that optimizes the system operation while respecting constraints and adjusting control variables, aiming to cover the optimization criteria that may arise in a real system through a broad range of objective functions.

PowerWorld Simulator (PWS). Includes specialized modules for load-flow solution, economic-dispatch analysis, fault calculation, sensitivity analysis, OPF, economic analysis of power transactions, transient-stability studies, contingency analysis, and short circuits, among others. Its optional OPF add-on optimally dispatches generation and computes the marginal price for the dispatch while satisfying constraints and accounting for system congestion.

Power System Simulator for Engineering (PSS/E). A high-performance suite offered by Siemens for steady-state and dynamic analysis and planning of power systems. It provides accurate results in power-flow studies, contingency analysis, OPF, and fault analysis, among others. Its OPF tool adds intelligence to the conventional flow solution, increasing performance and efficiency, and includes spreadsheets that let the user enter, modify, and analyze control variables and constraints.

As shown, several tools exist for OPF, classified as open-source or commercial. The open-source tools are the most accessible and the most extensively developed, and they focus specifically on OPF, whereas the specialized commercial tools focus on simulating EPS behavior—so solving optimization problems is not their sole objective. Reference [23] describes further open-source tools: for example, PSAT (Matlab/Simulink), used to evaluate dynamic networks; Dome, a Python-based tool derived from PSAT; GridCal, used for short-circuit calculations and power-flow analysis; and GridLAB-D, one of the most advanced C/C++ tools, integrated with SCADA controls and using an agent to simulate static models.

V. Conclusions

This report analyzed the importance of power flow and optimal power flow (OPF) in the operation and study of electric power systems. In general terms, OPF represents a modeling of the economic-dispatch problem based on the power-flow analysis of a system.

OPF can be structured in various ways depending on the optimization criterion adopted. Although its most common application is cost minimization, it can also target loss reduction, efficiency maximization, or improved system stability. This suggests that the key factor in the formulation is not only the objective function but the approach adopted to model the power flows and solve the problem.

Three main OPF models were reviewed: the lossless DC model, the DC model with losses, and the AC model. The lossless DC model is widely used in research and projects because of its simplified structure and lower computational demand, providing reasonably accurate approximations through linearizing assumptions. The DC model with losses adds complexity by considering network losses, yielding more realistic results without greatly compromising computational efficiency. The AC model is the most complete and accurate, since it considers both active and reactive power, which implies a more detailed formulation through Newton–Raphson and the system admittance matrix; although more demanding, it is fundamental for larger, deeper studies. Each model represents a trade-off between accuracy and computational complexity: AC OPF is ideal for detailed studies, while DC models are used in more general or economic analyses.

Given this complexity, several computational tools have been developed for implementation and solution. Two main categories were identified: commercial tools such as PowerFactory and ETAP, and open-source, programmable tools such as Matpower, Pypower, and Pandapower. The latter are especially useful in academic settings because of their accessibility and flexibility, and because they include widely used reference models.

OPF analysis is therefore essential for planning and operating power systems and for optimizing resources. The choice of model and solution tool depends on the required accuracy, the objective, and the available computational resources. Several studies proposing new OPF solution models were reviewed, which means that, as power systems evolve, the development of more sophisticated and accurate models will remain a key research topic.

VI. Recommendations

After this literature review, the following recommendations are offered:

- Although this work summarizes the essence and formulation of the OPF models, each cited study offers a distinct analytical perspective. Reviewing them in detail is recommended to broaden understanding and explore new perspectives.
- Any development of OPF models requires prior knowledge of conventional power flows. Reviewing the theoretical principles and associated processes is recommended, with special emphasis on the Newton–Raphson method, as it is the most used in these studies.
- Since power-flow studies usually involve tedious iterative analyses, using computational tools is advised, as they help visualize the processes and understand the logic behind each method and model.
- To deepen the study of power systems, specialized software is fundamental. Exploring environments such as MATLAB, Python, and DIgSILENT PowerFactory is recommended, as they facilitate the implementation and analysis of models.
- OPF is a broad field with multiple approaches. The best strategy to master it is to combine documentation review with practice through simulations, tests, and different scenarios. When direct programming is not possible, reading technical documentation remains a valuable tool for understanding OPF.

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